S · MITE

GENERAL ARRANGEMENT — SWARM UNIT

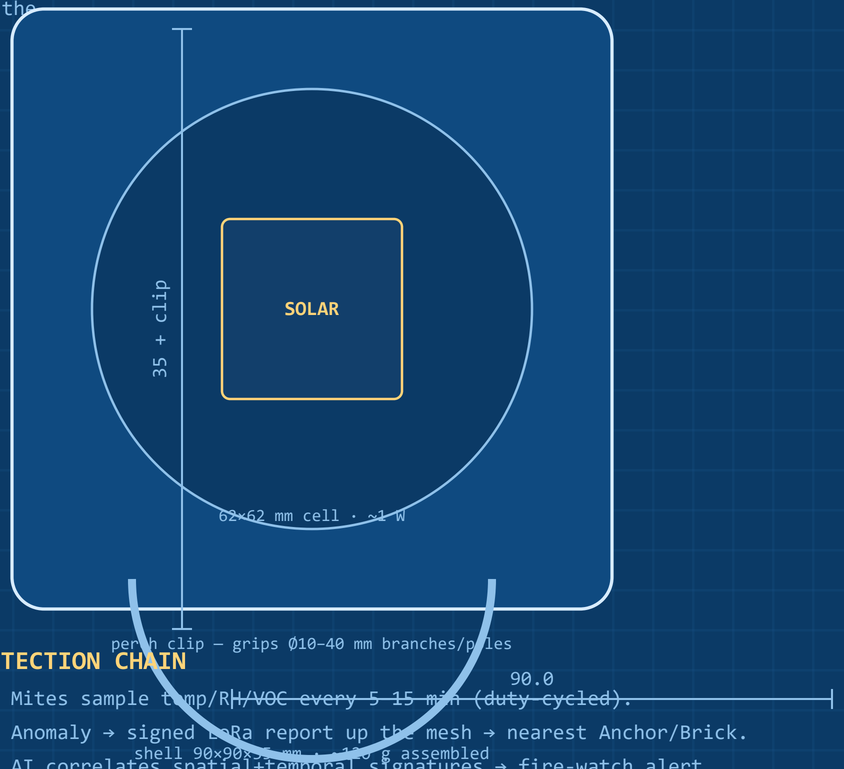
PERCHABLE ENVIRONMENTAL SENSOR NODE

AF-S-101
REV A · 2026-11-12
PROTOTYPE

DESIGN INTENT

- The swarm IS the sensor — a hundred perched Miters form a distributed aperture that follows the fire season.
- Each unit is deliberately minimal: solar top, LoRa radio, temp/RH/VOC sensing, perch clip. Roles are assigned by the swarm AI, not fixed in hardware.
- Default state = perched and reporting. Flight is a later role, not the power budget's problem.
- Same signed mesh identity as every ArcticChain node — a Mite is a node with a clip, not a separate system.

ENVELOPE — PERCH PUCK


35 + clip
SOLAR
62×62 mm cell · ~1 W
perch clip — grips Ø10–40 mm branches/poles
90.0
shell 90×90×35 mm · ~120 g assembled

SWARM ROLES (ASSIGNED BY MESH AI)

Role	Payload	Job
[illegible] primary detector		
[illegible] confirm		
[illegible] edges		
[illegible] or, top off, rebroadcast		

DETECTION CHAIN

- Mites sample temp/RH/VOC every 5–15 min (duty-cycled).
- Anomaly → signed LoRa report up the mesh → nearest Anchor/Brick.
- AI correlates spatial+temporal signatures → fire-watch alert.
- Eye-bees (later) converge on the fix for visual/thermal confirm.
- Weak distributed readings triangulate plumes better than one strong sensor.

GEN-1 vs ROADMAP

Stage	What	Gate
[illegible] perch clip		
[illegible] antenna		
[illegible] see D SEED		

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DRAWN: Surafel Negussie — Planner / Maker
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DWG AF-S-101
REV A
SCALE NTS · SHEET 1/4
DESIGNED FOR RURAL & REMOTE CANADA — -45..+70 °C · IP65 · SOLAR-FIRST · COMMUNITY-OWNED

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MECHANICAL & MOUNTING

PUCK SHELL, LID, BATTERY BAY, PERCH CLIP

AF-S-102

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PROTOTYPE

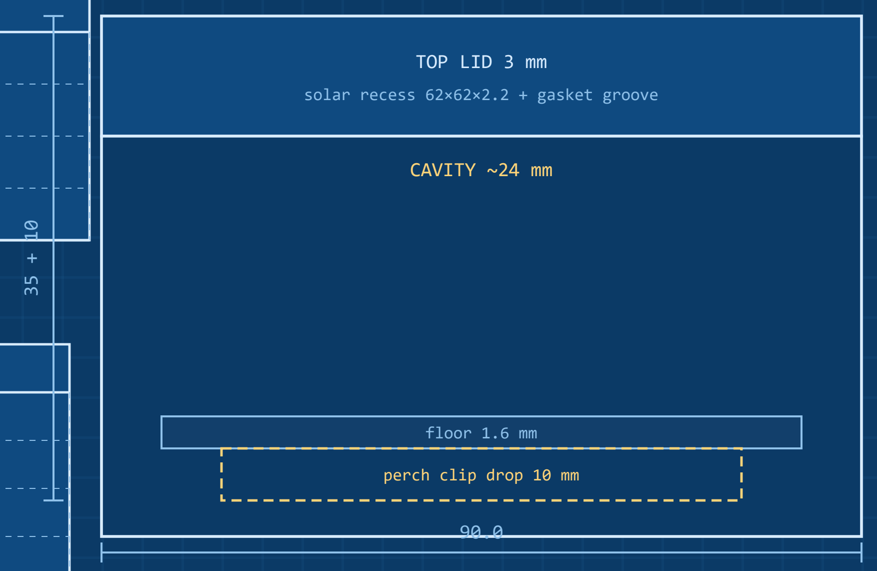
PARTS (cad/swarm_node.py → cad/swarm/*.stl)

Part	File	Material	Print
[illegible] (UV-sold)			
[illegible] solar recess			
[illegible]_bay.stl			
[illegible]			

ENVELOPE DIMENSIONS

Parameter	Value	Parameter	Value
[illegible] mm			
[illegible] 3/0 lid			
[illegible] (503450 LiPo)			
[illegible] body			
[illegible] silicone bead			

SECTION — SIDE VIEW (SCHEMATIC)



shell cavity fits: carrier PCB 38×38 + battery 54×38×9 + standoffs

MOUNTING — PERCH & GO

- Branch/pole: C-clip cradle grips Ø10–40 mm; node sits horizontal, solar up.
- Fence/wire: same clip works on barbed wire or T-posts.
- Metal surface: N52 magnet puck variant (rev-B) for barns, tanks, towers.
- Anchor dock: bottom pogo pads mate with Anchor charging contacts.
- No tools, no screws — deploy in seconds, relocate as fire risk moves.

WEATHERPROOFING

- Lid gasket groove + silicone bead → IP65 perched in rain/snow.
- Solar panel under optical window film (replaceable if abraded).
- Breather vent (ePTFE) optional rev-B for condensation in freeze-thaw.
- All fasteners captive — nothing to lose in the bush.

MATERIALS RATIONALE

- ASA/PC-ABS: UV-stable + tough to -45 °C (PETG gets brittle cold).
- Solar: 60×60 mm poly ~1 W — orders of magnitude over duty-cycled need.
- Clip: printed cradle + replaceable steel spring strip (field-swappable).
- No metal fasteners exposed — clip and lid are tool-free.

REV NOTES

- REV A: 90 mm puck, 503450 LiPo, manual assembly.
- REV B: magnet base variant, ePTFE breather, castellated battery.
- REV C: eye-bee payload bay (thermal cam / extra sensor).
- REV D: sub-bee mote — separate D SEED program, not a shrink of this.

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ELECTRICAL — BOM, POWER & PINOUT

PARTS + POWER BUDGET

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PROTOTYPE

BILL OF MATERIALS (GEN-1 — ALL ORDERABLE IN CANADA)

#	Part	Supplier class	Approx CAD	Qty
[illegible] module carrier				
[illegible] LoRa breakout 915 MHz				
[illegible] VOC/AIQ sensor (I2C)				
[illegible] temp/RH (I2C, optional 2nd)				
[illegible]				
[illegible] JST-PH leads				
[illegible] module (or TP4056-class)				
[illegible] mm (JLCPCB 5 pc)				
[illegible] spring strip				

POWER BUDGET (ESTIMATES — VERIFY ON BENCH)

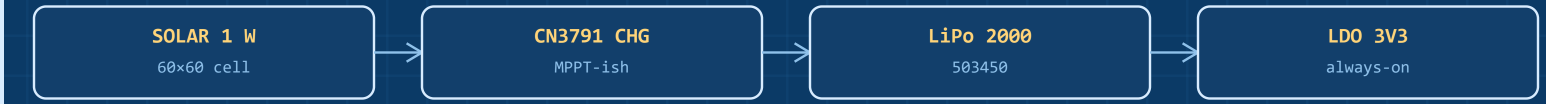
State	Current	Duty	Energy/day
[illegible] (ESP32)			
[illegible]			
[illegible] (30 dBm)			
[illegible] windows			
[illegible] Wh/day			

2000 mAh LiPo ≈ 7.4 Wh → ~200+ days theoretical; winter sun + self-discharge → plan 60–90 day autonomy.

CARRIER PCB PIN MAP (ESP32-S3 ↔ SX1262 ↔ SENSORS)

Net	ESP32-S3 GPIO	Function
[illegible] SPI clock		
[illegible]		
[illegible]		
[illegible] select		
[illegible]		
[illegible]/TX done IRQ		
[illegible] (0x53/0x44)		
[illegible] sense		
[illegible] detect		
[illegible] LED input		

SYSTEM BLOCK



ESP32-S3 + SX1262 + ENS160 on 38×38 carrier; all sensors on I2C; radio on private SPI bus.

ALTERNATES / FUTURE REVS

- Battery alt: 2×16340 LiPo (more compact) or single 18650 cold-rated (rev-B).
- Radio alt: SX1268 for +22 dBm if link budget demands (same footprint family).
- Sensor alt: SGP30 (cheaper VOC) or add SCD41 CO_2 for fire-season respiration proxy.
- Comms alt: LoRa backscatter / wake-up receiver for µW motes — D SEED research.

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FIRMWARE, ASSEMBLY & TEST

SOFTWARE ROLE + BUILD PROCEDURE

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PROTOTYPE

FIRMWARE ARCHITECTURE

Component	Detail
Runtime	ESP-IDF (C) or Rust-embedded — gen-1 C for speed; Rust port when mesh-net envelope lands
Envelope	Same signed format as mesh-net (Ed25519 + ChaCha20-Poly1305) — MUST match fleet
Duty-cycle	Deep-sleep 95% · sample 5 min · TX on anomaly or schedule · RX windows synced to mesh beacon
Role	Announces capabilities; swarm AI assigns sensor/eye/relay/charger roles via mesh messages
OTA	Signed firmware updates over mesh (rev-C); SD/USB for bench flashing
Data	Temp/RH/VOC + battery + RSSI + role → compressed observation frame (<200 B)

ASSEMBLY SEQUENCE

① Print shell, lid, battery bay, clip — ASA/PC-ABS shell, PETG clip strip.

② Populate carrier PCB: ESP32-S3, SX1262, ENS160, SHT41, charge circuit — or solder dev modules to carrier.

③ Seat battery in tray, connect JST; verify charge path with CN3791 (solar simulator or bench supply).

④ Close lid — gasket bead, solar cell under window film, captive screws/snaps.

⑤ Attach perch clip; bench-test: boot, join test mesh, confirm signed envelope round-trip.

⑥ Field trial: perch on fence line ×3 units, log range/RSSI/battery over 7 days.

ACCEPTANCE / TEST CRITERIA

- Two Miters exchange signed observations through a Brick/Anchor at ≥3 km LOS.
- Battery autonomy ≥60 days at 1 sample/10 min + hourly beacon (winter light derate).
- Survives -30 °C storage + operation; solar resumes charge at -10 °C (LiPo limit noted).
- Perch clip holds Ø25 mm branch through 60 km/h wind gust equivalent (shake test).
- False-positive smoke rate <1/week/unit in dust/inversion conditions (AI filter).

ROLE ASSIGNMENT PROTOCOL (SKETCH)

- All Miters broadcast hello + battery + capability every N hours.
- Anchor/Brick AI computes coverage gaps → assigns roles via signed cmd.
- Sensor-bees concentrate near fuel breaks/ridgelines in fire season.
- Charger-bees migrate to Anchor docks when battery <20%.
- Roles are soft state — units self-organize if the coordinator is gone.

OPEN ITEMS

- Carrier PCB layout — needs KiCad pass (this sheet = spec).
- ENS160 vs SGP30 selection on real smoke/dust test data.
- LoRa region config: 915 MHz ISM CA — duty-cycle limits apply.
- mesh-net envelope port to ESP32 (Rust-embedded or C shim).

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